Coverage Metric - GPU Optimized

Karina Arichavala

2026-05-07

Table of contents


```python
# Configuración para visualización en PDF
import pandas as pd
import numpy as np
import sys

# Para que el texto de pandas no se corte
pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)  # Ancho fijo para pandas

# Para prints largos de numpy - ancho más conservador
np.set_printoptions(linewidth=100, edgeitems=3)

# Configurar ancho de terminal para prints
import os
os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

```
 Configuración para PDF lista
```

0.1 1. Setup e Imports

```python
import numpy as np
import torch
import sys
import os
from pathlib import Path
from tqdm import tqdm
import time

# Configurar paths
project_root = Path('../../..').resolve()
sys.path.insert(0, str(project_root))
```

```python
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id
from sae.tools.experiment_utils import get_experiment_dir, get_metrics_dir, get_report_name

# Verificar GPU
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria disponible: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria disponible: 8.55 GB
```

```python
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',                 # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 1000,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',  # Path base para checkpoints
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments/
    'extras': {
        'expansion': 32,
        'sparsity_coef': 1e-2,
        'epochs': 250,
        'patience': 20,
        'learning_rate': 1e-3,
        'sparsity_decay': 0.98,
        'batch_size': 256
    }
}
print('\n Metadata del experimento:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata del experimento:
  variante: standard
  tecnica: l1-decay
  capa: 6
  juegos: 1000
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 32, 'sparsity_coef': 0.01, 'epochs': 250, 'patience': 20, 'learning_rate': 0.001, 'sparsity_dec
```

0.2 2. Cargar Datos

```python
# Cargar activaciones pre-SAE
activations_path = project_root / "sae" / "activations" / "data" / "layer6_1games.npy"
activations_full = np.load(activations_path)

activations = activations_full[-15:]

print(f"Activaciones del modelo:")
print(f"  Shape original: {activations_full.shape}")
print(f"  Shape final: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```
Activaciones del modelo:
  Shape original: (59, 512)
  Shape final: (15, 512)
  Memoria: 0.03 MB
```

```python
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_1games.npy"
bsp_names_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_1games.names.npy"

bsp_ground_truth_full = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)

bsp_ground_truth = bsp_ground_truth_full[-15:]
print(f"\nGround truth BSPs:")
print(f"  Shape original: {bsp_ground_truth_full.shape}")
print(f"  Shape final: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape original: (59, 326)
  Shape final: (15, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```python
# Configuración del SAE
input_dim = 512
hidden_dim = 16384

# Construir ruta del modelo usando naming_utils
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_name = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'best',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_name

# Cargar modelo (weights_only=False: checkpoint contiene objetos Path en config)
sae = SparseAutoencoder(input_dim, hidden_dim).to(device)
checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}")
print(f"  Expansion: {hidden_dim/input_dim}x")
print(f"  Epoch: {checkpoint.get('epoch', 'N/A')}, Val MSE: {checkpoint.get('val_mse', float('nan')):.6f}")
```

```
SAE cargado:
  Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\sae_l1-decay_standard\1000games\sa
decay_l6_1000g_ex5_best.pt
  Input: 512, Hidden: 16384
  Expansion: 32.0x
  Epoch: N/A, Val MSE: nan
```

```python
# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

with torch.no_grad():
    sae_features = torch.relu(sae.encoder(activations_tensor))

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

```
Extrayendo features del SAE...

Features SAE:
  Shape: torch.Size([15, 16384])
  Device: cuda:0
  Sparsity: 99.83%
  Activaciones promedio: 28.5
```

0.4 4. Filtrar BSPs de Piezas

Coverage solo usa las 128 BSPs de piezas (mías/oponente), sin vacías.

```python
# ============================================
# CONFIGURACIÓN DE FILTRADO DE BSPs
# ============================================
USE_PLAYER   = False   # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = True  # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False # BSPs especiales (BSP_)

# Filtrar según configuración
bsp_pieces_indices = []
bsp_pieces_names = []

for i, name in enumerate(bsp_names):
    include = False

    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True

    if include:
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)
bsp_pieces_gt = bsp_ground_truth[:, bsp_pieces_indices]

# Convertir a tensor en GPU
```

```python
bsp_pieces_gt_tensor = torch.from_numpy(bsp_pieces_gt).bool().to(device)

print(f"BSPs seleccionadas para Coverage:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Shape: {bsp_pieces_gt_tensor.shape}")
print(f"  Device: {bsp_pieces_gt_tensor.device}")
print(f"  Primeras 5: {', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {', '.join(bsp_pieces_names[-5:])}")
# Identificador del filtrado (para naming de archivos de salida)
parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"
print(f"  Filter suffix: {filter_suffix}")
```

```
BSPs seleccionadas para Coverage:
  Player: False | Absolute: True | Strategic: False
  Total: 128
  Shape: torch.Size([15, 128])
  Device: cuda:0
  Primeras 5: BSPA1B, BSPA1W, BSPA2B, BSPA2W, BSPA3B
  Últimas 5: BSPH6W, BSPH7B, BSPH7W, BSPH8B, BSPH8W
  Filter suffix: absolute
```

0.5 5. Implementación de Coverage

0.5.1 Estrategia de optimización:

1. **Vectorización**: Procesar múltiples features en paralelo usando operaciones matriciales
2. **Batch processing**: Dividir en chunks para no saturar memoria GPU
3. **Pre-cálculo**: Calcular máximos una sola vez
4. **Early stopping**: Terminar cuando F1   1.0

```python
def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        f1_scores: Tensor (n_candidates,)
    """
    # y_true: (n_positions,) -> expand to (n_positions, 1)
    y_true_expanded = y_true.unsqueeze(1)  # (n_positions, 1)

    # Calcular TP, FP, FN
    tp = (y_true_expanded & y_pred).sum(dim=0).float()  # (n_candidates,)
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()
    fn = (y_true_expanded & ~y_pred).sum(dim=0).float()

    # Precision y Recall
    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    # F1
    f1 = 2 * (precision * recall) / (precision + recall + eps)
```

```python
    return f1


def calculate_coverage_gpu_optimized(
    sae_features,  # Tensor (n_positions, n_features) en GPU
    bsp_ground_truth,  # Tensor (n_positions, n_bsps) en GPU, bool
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    batch_size_features=512,  # Procesar 512 features a la vez
    verbose=True
):
    """
    Calcula Coverage de forma optimizada para GPU.

    Procesa múltiples features y thresholds en paralelo usando operaciones matriciales.
    """
    n_positions, n_features = sae_features.shape
    n_bsps = bsp_ground_truth.shape[1]
    n_thresholds = len(thresholds)

    if verbose:
        print("="*60)
        print("CALCULATING COVERAGE (GPU OPTIMIZED)")
        print("="*60)
        print(f"Positions: {n_positions:,}")
        print(f"SAE Features: {n_features:,}")
        print(f"BSPs: {n_bsps}")
        print(f"Thresholds: {n_thresholds}")
        print(f"Batch size (features): {batch_size_features}")
        print(f"Total evaluations: {n_bsps * n_features * n_thresholds:,}")
        print("="*60)

    # Pre-calcular máximos de features
    f_max = sae_features.max(dim=0)[0]  # (n_features,)
    active_features = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_features)

    if verbose:
        print(f"\nActive features: {n_active:,}/{n_features:,} ({n_active/n_features*100:.1f}%)")
        print(f"Reduced evaluations: {n_bsps * n_active * n_thresholds:,}")
        print()

    # Almacenar resultados
    best_f1s = torch.zeros(n_bsps, device=sae_features.device)
    best_features = torch.full((n_bsps,), -1, dtype=torch.long, device=sae_features.device)
    best_thresholds = torch.full((n_bsps,), -1.0, device=sae_features.device)

    thresholds_tensor = torch.tensor(thresholds, device=sae_features.device)

    # Procesar cada BSP
    pbar = tqdm(range(n_bsps), desc="Processing BSPs") if verbose else range(n_bsps)

    for bsp_idx in pbar:
        bsp_labels = bsp_ground_truth[:, bsp_idx]  # (n_positions,)

        # Skip si la BSP nunca está activa
        if not bsp_labels.any():
            continue

        # Procesar features activas en batches
        for batch_start in range(0, n_active, batch_size_features):
            batch_end = min(batch_start + batch_size_features, n_active)
            batch_features_idx = active_features[batch_start:batch_end]
```

```
            # Extraer features del batch
            batch_activations = sae_features[:, batch_features_idx]  # (n_positions, batch_size)
            batch_f_max = f_max[batch_features_idx]  # (batch_size,)

            # Para cada threshold, binarizar TODAS las features del batch a la vez
            for t in thresholds_tensor:
                # Binarizar: (n_positions, batch_size)
                predictions = batch_activations > (t * batch_f_max.unsqueeze(0))

                # Calcular F1 para todas las features del batch
                f1_scores = fast_f1_score_gpu(bsp_labels, predictions)  # (batch_size,)

                # Encontrar la mejor en este batch
                max_f1, max_idx_in_batch = f1_scores.max(dim=0)

                # Actualizar si es mejor que lo visto hasta ahora
                if max_f1 > best_f1s[bsp_idx]:
                    best_f1s[bsp_idx] = max_f1
                    best_features[bsp_idx] = batch_features_idx[max_idx_in_batch]
                    best_thresholds[bsp_idx] = t

            # Early stopping: si ya es casi perfecto, no revisar más batches
            if best_f1s[bsp_idx] >= 0.999:
                break

    # Calcular Coverage (macro-average)
    coverage = best_f1s.mean().item()

    # Convertir a CPU para retornar
    best_f1s_cpu = best_f1s.cpu().numpy()
    best_features_cpu = best_features.cpu().numpy()
    best_thresholds_cpu = best_thresholds.cpu().numpy()

    return coverage, best_f1s_cpu, best_features_cpu, best_thresholds_cpu

print(" Funciones de Coverage GPU definidas")
```

```
 Funciones de Coverage GPU definidas
```

0.6 6. Calcular Coverage

```
# Medir tiempo de ejecución
start_time = time.time()

coverage, best_f1s, best_features, best_thresholds = calculate_coverage_gpu_optimized(
    sae_features,
    bsp_pieces_gt_tensor,
    batch_size_features=512,  # Ajustar según memoria GPU
    verbose=True
)

elapsed_time = time.time() - start_time

print("\n" + "="*60)
print("RESULTADO COVERAGE")
print("="*60)
print(f"Coverage Score: {coverage:.4f}")
print(f"Objetivo (paper): 0.52")
print(f"Diferencia: {coverage - 0.52:+.4f}")
```

```python
print(f"\nTiempo de ejecución: {elapsed_time:.2f} seg")
print(f"                      ({elapsed_time/60:.2f} min)")
print("="*60)
```

```
============================================================
CALCULATING COVERAGE (GPU OPTIMIZED)
============================================================
Positions: 15
SAE Features: 16,384
BSPs: 128
Thresholds: 10
Batch size (features): 512
Total evaluations: 20,971,520
============================================================

Active features: 29/16,384 (0.2%)
Reduced evaluations: 37,120

Processing BSPs: 100%|     | 128/128 [00:00<00:00, 150.79it/s]


============================================================
RESULTADO COVERAGE
============================================================
Coverage Score: 0.6757
Objetivo (paper): 0.52
Diferencia: +0.1557

Tiempo de ejecución: 0.89 seg
                      (0.01 min)
============================================================
```

0.7 7. Análisis de Resultados

```python
# Estadísticas de distribución de F1 scores
print("Distribución de F1 scores:")
print(f"  Mínimo: {best_f1s.min():.4f}")
print(f"  Máximo: {best_f1s.max():.4f}")
print(f"  Media: {best_f1s.mean():.4f}")
print(f"  Mediana: {np.median(best_f1s):.4f}")
print(f"  Std: {best_f1s.std():.4f}")
print()
print(f"BSPs con F1 > 0.8: {(best_f1s > 0.8).sum()} ({(best_f1s > 0.8).mean()*100:.1f}%)")
print(f"BSPs con F1 > 0.5: {(best_f1s > 0.5).sum()} ({(best_f1s > 0.5).mean()*100:.1f}%)")
print(f"BSPs con F1 < 0.2: {(best_f1s < 0.2).sum()} ({(best_f1s < 0.2).mean()*100:.1f}%)")
```

```
Distribución de F1 scores:
  Mínimo: 0.0000
  Máximo: 1.0000
  Media: 0.6757
  Mediana: 0.9045
  Std: 0.4150

BSPs con F1 > 0.8: 80 (62.5%)
BSPs con F1 > 0.5: 94 (73.4%)
BSPs con F1 < 0.2: 34 (26.6%)
```

```python
# Top 10 BSPs mejor detectadas
top_indices = np.argsort(best_f1s)[-10:][::-1]

print("\nTop 10 BSPs mejor detectadas:")
print("="*60)
print(f"{'BSP':<10} {'F1':<8} {'Feature':<10} {'Threshold':<10}")
print("-"*60)
for idx in top_indices:
    bsp_name = bsp_pieces_names[idx]
    f1 = best_f1s[idx]
    feat = best_features[idx]
    thresh = best_thresholds[idx]
    print(f"{bsp_name:<10} {f1:<8.4f} {feat:<10} {thresh:<10.1f}")
```

```
Top 10 BSPs mejor detectadas:
============================================================
BSP        F1       Feature    Threshold
------------------------------------------------------------
BSPH8W     1.0000   83         0.0
BSPH6B     1.0000   83         0.0
BSPH5B     1.0000   83         0.0
BSPH4W     1.0000   83         0.0
BSPH2W     1.0000   12867      0.2
BSPH1W     1.0000   83         0.0
BSPH3W     1.0000   12867      0.2
BSPG8W     1.0000   83         0.0
BSPG6B     1.0000   83         0.0
BSPG5W     1.0000   83         0.0
```

```python
# Bottom 10 BSPs peor detectadas
bottom_indices = np.argsort(best_f1s)[:10]

print("\nBottom 10 BSPs peor detectadas:")
print("="*60)
print(f"{'BSP':<10} {'F1':<8} {'Feature':<10} {'Threshold':<10}")
print("-"*60)
for idx in bottom_indices:
    bsp_name = bsp_pieces_names[idx]
    f1 = best_f1s[idx]
    feat = best_features[idx]
    thresh = best_thresholds[idx]
    print(f"{bsp_name:<10} {f1:<8.4f} {feat:<10} {thresh:<10.1f}")
```

```
Bottom 10 BSPs peor detectadas:
============================================================
BSP        F1       Feature    Threshold
------------------------------------------------------------
BSPA1W     0.0000   -1         -1.0
BSPA2W     0.0000   -1         -1.0
BSPA3B     0.0000   -1         -1.0
BSPA8W     0.0000   -1         -1.0
BSPA7B     0.0000   -1         -1.0
BSPB8W     0.0000   -1         -1.0
BSPD7B     0.0000   -1         -1.0
BSPC8W     0.0000   -1         -1.0
BSPD1W     0.0000   -1         -1.0
BSPD8B     0.0000   -1         -1.0
```

```python
# Distribución de thresholds óptimos
unique_thresholds, counts = np.unique(best_thresholds, return_counts=True)

print("\nDistribución de thresholds óptimos:")
print("="*60)
for t, count in zip(unique_thresholds, counts):
    if t >= 0:  # Ignorar -1 (BSPs sin match)
        percentage = count / len(best_thresholds) * 100
        print(f"Threshold {t:.1f}: {count:3d} BSPs ({percentage:.1f}%)")
```

```
Distribución de thresholds óptimos:
============================================================
Threshold 0.0:  25 BSPs (19.5%)
Threshold 0.1:   5 BSPs (3.9%)
Threshold 0.2:  15 BSPs (11.7%)
Threshold 0.3:   5 BSPs (3.9%)
Threshold 0.4:  10 BSPs (7.8%)
Threshold 0.5:   5 BSPs (3.9%)
Threshold 0.6:   3 BSPs (2.3%)
Threshold 0.7:  12 BSPs (9.4%)
Threshold 0.8:   9 BSPs (7.0%)
Threshold 0.9:   5 BSPs (3.9%)
```

```python
# ============================================
# CONSULTA POR BSP ESPECÍFICA
# ============================================

# >>> CAMBIA AQUÍ LA BSP QUE QUIERES ANALIZAR <
BSP_OBJETIVO = "BSPG5W"

# Buscar índice de la BSP
bsp_idx = bsp_pieces_names.index(BSP_OBJETIVO)

# Extraer resultados ya calculados
neurona   = best_features[bsp_idx]
threshold = best_thresholds[bsp_idx]
f1        = best_f1s[bsp_idx]

# f_max de la neurona ganadora (sae_features ya está en memoria)
f_max_neurona = sae_features[:, neurona].max().item()

# Activaciones de esa neurona en todos los tableros
activaciones_neurona = sae_features[:, neurona].cpu().numpy()

print(f"BSP analizada:     {BSP_OBJETIVO}")
print(f"Neurona ganadora:  {neurona}")
print(f"Threshold óptimo:  {threshold:.1f}")
print(f"F_max neurona:     {f_max_neurona:.4f}")
print(f"Umbral efectivo:   {threshold:.1f} x {f_max_neurona:.4f} = {threshold * f_max_neurona:.4f}")
print(f"F1 score:          {f1:.4f}")
print(f"\nActivaciones de neurona {neurona} ({len(activaciones_neurona)} tableros):")
for i, act in enumerate(activaciones_neurona):
    print(f"  Tablero {i+1:5d}: {act:.6f}")
```

```
BSP analizada:     BSPG5W
Neurona ganadora:  83
Threshold óptimo:  0.0
F_max neurona:     0.2469
Umbral efectivo:   0.0 x 0.2469 = 0.0000
F1 score:          1.0000
```

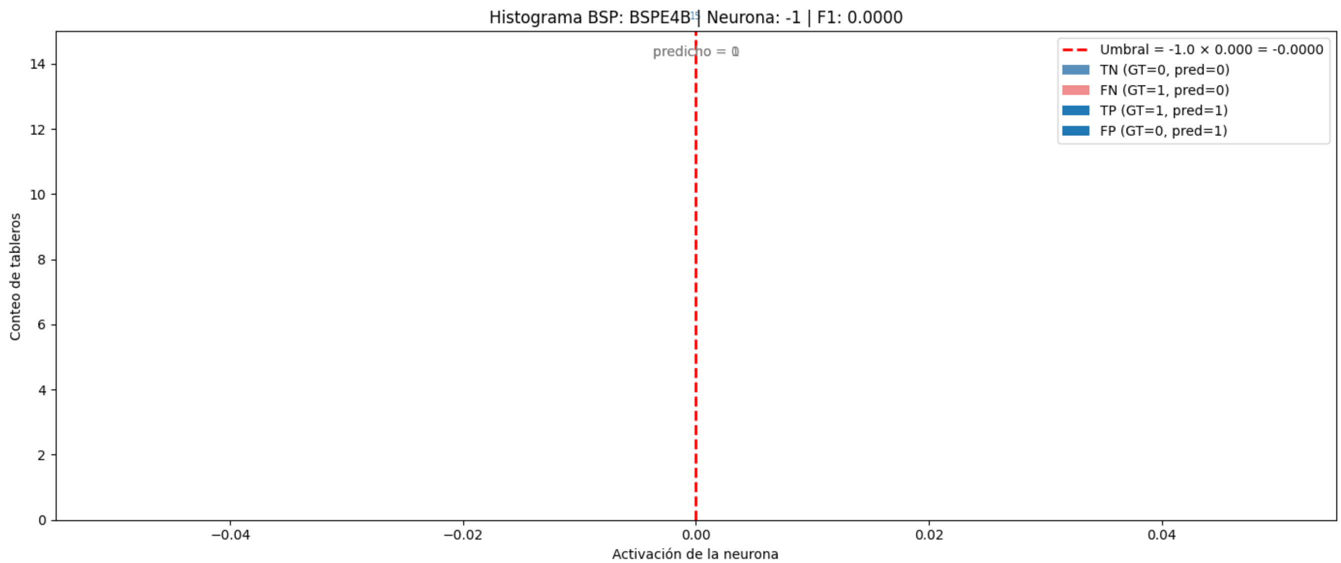
```
Activaciones de neurona 83 (15 tableros):
  Tablero     1: 0.246944
  Tablero     2: 0.194055
  Tablero     3: 0.178463
  Tablero     4: 0.142503
  Tablero     5: 0.151989
  Tablero     6: 0.171002
  Tablero     7: 0.237638
  Tablero     8: 0.201972
  Tablero     9: 0.206116
  Tablero    10: 0.179602
  Tablero    11: 0.201458
  Tablero    12: 0.140281
  Tablero    13: 0.171347
  Tablero    14: 0.206104
  Tablero    15: 0.133206
```

```
def plot_bsp_histogram(BSP_HISTOGRAMA, sae_features, bsp_pieces_names, bsp_pieces_gt_tensor,
                       best_features, best_thresholds, best_f1s, n_bins=20):

    # Extraer datos de la BSP
    bsp_idx_hist   = bsp_pieces_names.index(BSP_HISTOGRAMA)
    neurona_hist   = best_features[bsp_idx_hist]
    threshold_hist = best_thresholds[bsp_idx_hist]
    f1_hist        = best_f1s[bsp_idx_hist]
    f_max_hist     = sae_features[:, neurona_hist].max().item()

    # Activaciones y ground truth
    activaciones = sae_features[:, neurona_hist].cpu().numpy()
    gt           = bsp_pieces_gt_tensor[:, bsp_idx_hist].cpu().numpy()
    umbral       = threshold_hist * f_max_hist

    # Clasificar
    predicho = (activaciones > umbral).astype(int)
    tp_mask  = (gt == 1) & (predicho == 1)
    fp_mask  = (gt == 0) & (predicho == 1)
    tn_mask  = (gt == 0) & (predicho == 0)
    fn_mask  = (gt == 1) & (predicho == 0)

    # Bins
    bins        = np.linspace(0, activaciones.max(), n_bins + 1)
    bin_centers = (bins[:-1] + bins[1:]) / 2
    bin_width   = bins[1] - bins[0]

    tp_counts, _ = np.histogram(activaciones[tp_mask], bins=bins)
    fp_counts, _ = np.histogram(activaciones[fp_mask], bins=bins)
    tn_counts, _ = np.histogram(activaciones[tn_mask], bins=bins)
    fn_counts, _ = np.histogram(activaciones[fn_mask], bins=bins)

    left_mask  = bin_centers <= umbral
    right_mask = bin_centers > umbral

    # Graficar
    fig, ax = plt.subplots(figsize=(14, 6))

    ax.bar(bin_centers[left_mask], tn_counts[left_mask],
           width=bin_width, color='steelblue', alpha=0.9,
           label='TN (GT=0, pred=0)', align='center')
    ax.bar(bin_centers[left_mask], fn_counts[left_mask],
           width=bin_width, bottom=tn_counts[left_mask],
           color='lightcoral', alpha=0.9,
```

```python
            label='FN (GT=1, pred=0)', align='center')
    ax.bar(bin_centers[right_mask], tp_counts[right_mask],
           width=bin_width, color='green', alpha=0.9,
           label='TP (GT=1, pred=1)', align='center')
    ax.bar(bin_centers[right_mask], fp_counts[right_mask],
           width=bin_width, bottom=tp_counts[right_mask],
           color='orange', alpha=0.9,
           label='FP (GT=0, pred=1)', align='center')

    # Números TN
    for i, x in enumerate(bin_centers[left_mask]):
        total = tn_counts[left_mask][i] + fn_counts[left_mask][i]
        if tn_counts[left_mask][i] > 0:
            ax.text(x, total + 0.3, str(tn_counts[left_mask][i]),
                    ha='center', va='bottom', fontsize=7, color='steelblue')

    # Números TP
    for i, x in enumerate(bin_centers[right_mask]):
        total = tp_counts[right_mask][i] + fp_counts[right_mask][i]
        if tp_counts[right_mask][i] > 0:
            ax.text(x, total + 0.3, str(tp_counts[right_mask][i]),
                    ha='center', va='bottom', fontsize=7, color='green')

    # Umbral
    ax.axvline(x=umbral, color='red', linewidth=2, linestyle='--',
               label=f'Umbral = {threshold_hist:.1f} × {f_max_hist:.3f} = {umbral:.4f}')

    ax.set_xlabel('Activación de la neurona')
    ax.set_ylabel('Conteo de tableros')
    ax.set_title(f'Histograma BSP: {BSP_HISTOGRAMA} | Neurona: {neurona_hist} | F1: {f1_hist:.4f}')
    ax.legend()
    ax.text(umbral * 0.5, ax.get_ylim()[1] * 0.95, 'predicho = 0',
            ha='center', fontsize=10, color='gray')
    ax.text(umbral + (activaciones.max() - umbral) * 0.5, ax.get_ylim()[1] * 0.95, 'predicho = 1',
            ha='center', fontsize=10, color='gray')

    plt.tight_layout()
    plt.show()

    print(f"\nResumen:")
    print(f"  TP: {tp_mask.sum()} | FP: {fp_mask.sum()}")
    print(f"  TN: {tn_mask.sum()} | FN: {fn_mask.sum()}")
    print(f"  F1: {f1_hist:.4f}")

print("  Función plot_bsp_histogram definida")
```

```
  Función plot_bsp_histogram definida
```

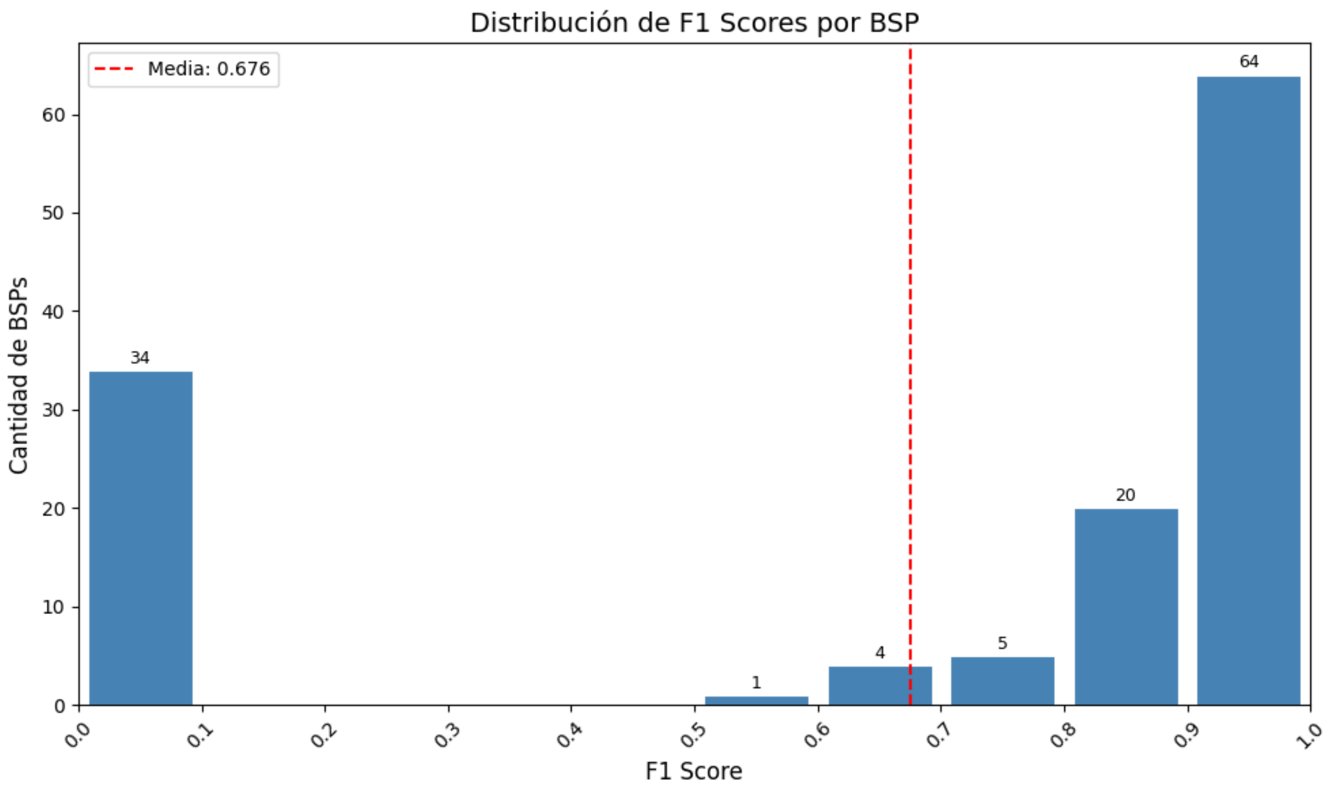
```python
# >>> CAMBIA AQUÍ LA BSP QUE QUIERES VISUALIZAR <
import matplotlib.pyplot as plt
plot_bsp_histogram(
    BSP_HISTOGRAMA       = "BSPE4B",
    sae_features         = sae_features,
    bsp_pieces_names     = bsp_pieces_names,
    bsp_pieces_gt_tensor = bsp_pieces_gt_tensor,
    best_features        = best_features,
    best_thresholds      = best_thresholds,
    best_f1s             = best_f1s,
    n_bins               = 10
)
```




```
Resumen:
  TP: 0 | FP: 0
  TN: 15 | FN: 0
  F1: 0.0000
```

```python
import matplotlib.pyplot as plt
import numpy as np

def plot_f1_distribution(best_f1s, bsp_pieces_names, n_bins=10):
    """
    Gráfico 1: Distribución de F1 scores por BSP.

    Args:
        best_f1s: array numpy con el mejor F1 score de cada BSP
        bsp_pieces_names: lista con los nombres de las BSPs
        n_bins: número de bins del histograma
    """
    fig, ax = plt.subplots(figsize=(10, 6))

    bins = np.linspace(0, 1, n_bins + 1)
    counts, edges = np.histogram(best_f1s, bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2
    bin_width = edges[1] - edges[0]

    bars = ax.bar(
        bin_centers, counts,
        width=bin_width * 0.85,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
```

```
        )

    ax.set_xlabel('F1 Score', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de F1 Scores por BSP', fontsize=14)
    ax.set_xticks(bins)
    ax.set_xticklabels([f'{b:.1f}' for b in bins], rotation=45)
    ax.set_xlim(0, 1)

    # Línea de la media
    mean_f1 = best_f1s.mean()
    ax.axvline(mean_f1, color='red', linestyle='--', linewidth=1.5, label=f'Media: {mean_f1:.3f}')
    ax.legend(fontsize=10)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs: {len(best_f1s)}")
    print(f"Media F1: {mean_f1:.4f}")
    print(f"BSPs con F1 = 0: {(best_f1s == 0).sum()}")
    print(f"BSPs con F1 > 0.5: {(best_f1s > 0.5).sum()}")
    print(f"BSPs con F1 > 0.9: {(best_f1s > 0.9).sum()}")


# Ejecutar
plot_f1_distribution(best_f1s, bsp_pieces_names)
```

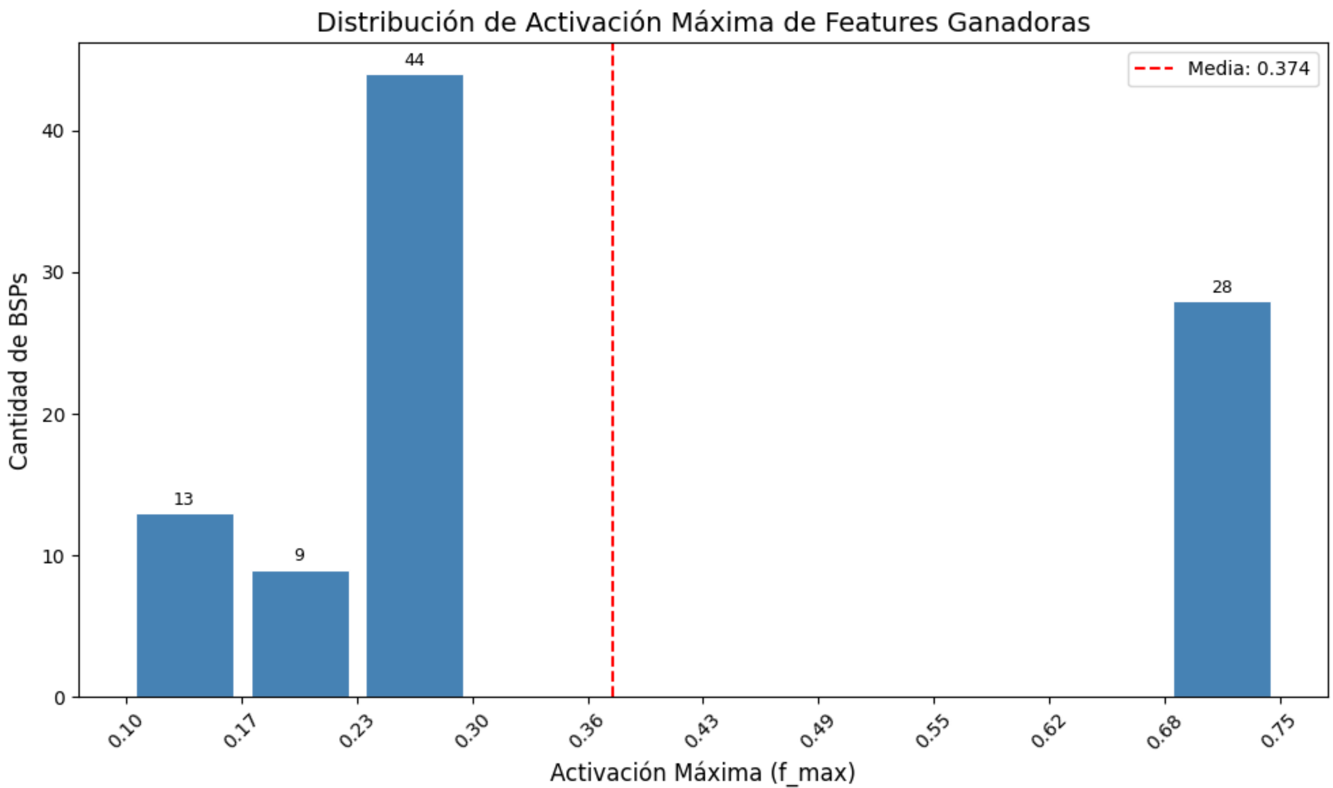



```
Total BSPs: 128
Media F1: 0.6757
BSPs con F1 = 0: 34
BSPs con F1 > 0.5: 94
BSPs con F1 > 0.9: 64
```

```python
# Extraer f_max de las features activas
f_max = sae_features.max(dim=0)[0]  # (n_features,)
```

```python
def plot_max_activation_distribution(best_features, f_max, bsp_pieces_names, n_bins=10):
    """
    Gráfico 2: Distribución de activación máxima de la feature ganadora por BSP.

    Args:
        best_features: array numpy con el índice de la feature ganadora de cada BSP
        f_max: tensor con la activación máxima de cada feature
        bsp_pieces_names: lista con los nombres de las BSPs
        n_bins: número de bins del histograma
    """
    # Obtener f_max de la feature ganadora de cada BSP
    # Ignorar BSPs sin feature ganadora (best_features == -1)
    valid_mask = best_features >= 0
    valid_features = best_features[valid_mask]
    f_max_winners = f_max[valid_features].cpu().numpy()

    fig, ax = plt.subplots(figsize=(10, 6))

    bins = np.linspace(f_max_winners.min(), f_max_winners.max(), n_bins + 1)
    counts, edges = np.histogram(f_max_winners, bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2
    bin_width = edges[1] - edges[0]

    bars = ax.bar(
        bin_centers, counts,
        width=bin_width * 0.85,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    # Línea de la media
    mean_fmax = f_max_winners.mean()
    ax.axvline(mean_fmax, color='red', linestyle='--', linewidth=1.5,
               label=f'Media: {mean_fmax:.3f}')
    ax.legend(fontsize=10)

    ax.set_xlabel('Activación Máxima (f_max)', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de Activación Máxima de Features Ganadoras', fontsize=14)
    ax.set_xticks(edges)
    ax.set_xticklabels([f'{b:.2f}' for b in edges], rotation=45)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs con feature ganadora: {valid_mask.sum()}")
    print(f"BSPs sin feature ganadora: {(~valid_mask).sum()}")
    print(f"Media f_max ganadora: {mean_fmax:.4f}")
```

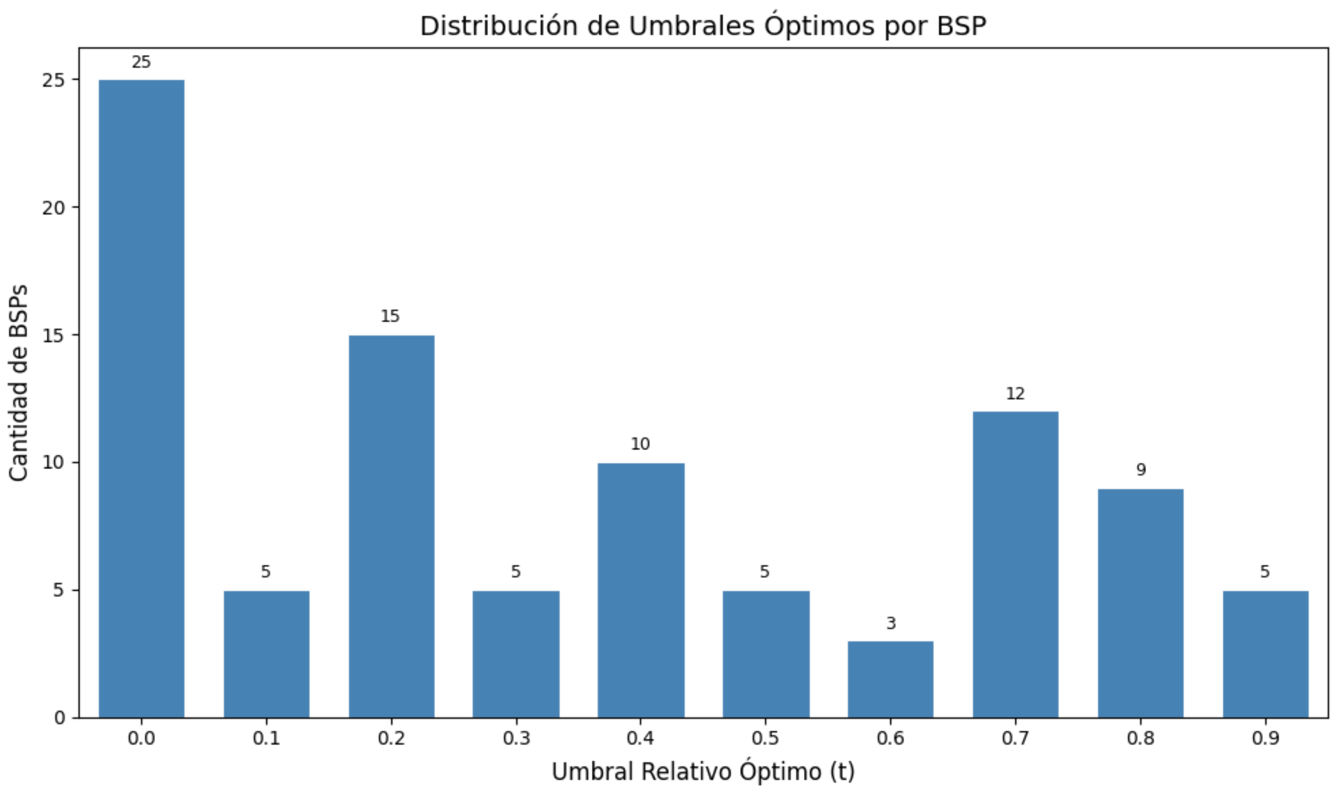
```
    print(f"Mínimo f_max: {f_max_winners.min():.4f}")
    print(f"Máximo f_max: {f_max_winners.max():.4f}")


# Ejecutar
plot_max_activation_distribution(best_features, f_max, bsp_pieces_names)
```




```
Total BSPs con feature ganadora: 94
BSPs sin feature ganadora: 34
Media f_max ganadora: 0.3743
Mínimo f_max: 0.1028
Máximo f_max: 0.7473
```

```
def plot_threshold_distribution(best_thresholds, bsp_pieces_names):
    """
    Gráfico 3: Distribución de umbrales óptimos por BSP.

    Args:
        best_thresholds: array numpy con el umbral óptimo de cada BSP
        bsp_pieces_names: lista con los nombres de las BSPs
    """
    # Ignorar BSPs sin feature ganadora (best_thresholds == -1)
    valid_mask = best_thresholds >= 0
    valid_thresholds = best_thresholds[valid_mask]

    fig, ax = plt.subplots(figsize=(10, 6))

    # Los umbrales son discretos: 0.0, 0.1, 0.2, ..., 0.9
    threshold_values = np.arange(0.0, 1.0, 0.1)
    counts = [(np.abs(valid_thresholds - t) < 0.05).sum() for t in threshold_values]

    bars = ax.bar(
        threshold_values, counts,
```

```
        width=0.07,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    ax.set_xlabel('Umbral Relativo Óptimo (t)', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de Umbrales Óptimos por BSP', fontsize=14)
    ax.set_xticks(threshold_values)
    ax.set_xticklabels([f'{t:.1f}' for t in threshold_values])
    ax.set_xlim(-0.05, 0.95)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs con umbral óptimo: {valid_mask.sum()}")
    print(f"BSPs sin feature ganadora: {(~valid_mask).sum()}")
    print(f"Umbral más frecuente: {threshold_values[np.argmax(counts)]:.1f}")


# Ejecutar
plot_threshold_distribution(best_thresholds, bsp_pieces_names)
```



```
Total BSPs con umbral óptimo: 94
BSPs sin feature ganadora: 34
Umbral más frecuente: 0.0
```

0.8 8. Guardar Resultados

```python
# Guardar resultados en el directorio de métricas del experimento
results = {
    'coverage': coverage,
    'best_f1s': best_f1s,
    'best_features': best_features,
    'best_thresholds': best_thresholds,
    'bsp_names': bsp_pieces_names,
    'execution_time_seconds': elapsed_time
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"coverage_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f" Resultados guardados en:")
print(f"  {output_path}")
```

```
 Resultados guardados en:
  c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_l1-decay_standard\1000games\e
```

0.9 9. Generar PDF con Quarto

```python
import subprocess

# Guardar PDF en el mismo directorio que el .npz (metrics/)
pdf_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    f'coverage_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'coverage_gpu_optimized.ipynb'
output_path = pdf_dir / pdf_name
```

```python
print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{pdf_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_l1-
decay_standard\1000games\ex5\metrics\sae_standard_l1-decay_l6_1000g_ex5_coverage_absolute.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_l1-
decay_standard\1000games\ex5\metrics\sae_standard_l1-decay_l6_1000g_ex5_coverage_absolute.pdf
```